

CORE

Brand Mention Tracking Platform

Technical Report

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Google Hackathon Project

Built with React · Vite · Tailwind CSS · Multi-LLM APIs · Python

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1. Executive Summary

CORE (Content Optimization & Ranking Engine) is a web-based platform designed to help businesses monitor, analyze, and improve how their brand is mentioned across major AI language models. As AI-powered search engines and conversational assistants (ChatGPT, Gemini, Claude, etc.) increasingly influence consumer decisions, brand visibility in these systems has become a critical marketing frontier. CORE provides a unified dashboard to track brand mentions, perform sentiment analysis, benchmark against competitors, and generate optimized content — all driven by multi-LLM intelligence.

This report details the system architecture, technology stack, AI integrations, key features, and future roadmap of the CORE platform, developed as part of the Google Hackathon 2026.

2. Project Objective

The primary objective of CORE is to provide businesses with actionable intelligence on how their brand is represented by AI language models. Specifically, the platform aims to:

- **Monitor brand mentions** across multiple LLMs (ChatGPT, Gemini, Claude, Mistral, DeepSeek) in real time.
- **Analyze sentiment** and ranking position of the brand within AI-generated responses.
- **Benchmark against competitors** to identify visibility gaps and opportunities.
- **Support geographic intelligence (GEO)** — tracking brand perception across countries and languages.
- **Automate monitoring** via scheduled scans (daily, weekly, custom intervals).
- **Generate optimized content** through an AI Article Engine that mass-produces SEO/GEO-optimized blog posts.
- **Publish content** to external endpoints to improve brand presence in AI-powered web search.

3. System Architecture

CORE follows a client-side single-page application (SPA) architecture with Python microservices for advanced backend tasks. The system is composed of three main layers:

3.1 Frontend (React SPA)

The entire user interface is built as a React SPA powered by Vite. The application uses React Router for client-side routing and Tailwind CSS for styling. Key pages include the landing page, onboarding wizard, user dashboard (brand management, query management, analysis execution, AI Article Engine, and results visualization), and a demo mode for quick evaluation.

3.2 Service Layer

Two primary service modules handle the business logic: **llmService.js** manages all API interactions with LLM providers (handling both OpenAI-compatible and Google Gemini API formats, parallel query execution, and rate limiting), while **analysisService.js** orchestrates brand analysis workflows

including mention detection, sentiment scoring, rank extraction, and URL discovery.

3.3 Python Backend Services

Python scripts provide supplementary capabilities: `get_context.py` uses Selenium and GPT-4o-mini to scrape and summarize company websites, while `get_trendy_keywords.py` leverages the Google Trends API (pytrends) to discover trending keywords relevant to the brand's industry.

4. Technology Stack

Layer	Technology	Version / Details
Frontend Framework	React	18.3.1
Build Tool	Vite	5.4.0
Styling	Tailwind CSS	3.4.10
Routing	React Router DOM	6.26.1
PDF Export	jsPDF + html2canvas	3.0.3 / 1.4.1
CSS Processing	PostCSS + Autoprefixer	8.4.45 / 10.4.19
Language	JavaScript (ES2022+)	—
Python Runtime	Python	3.11+
Web Scraping	Selenium	4.41.0
AI Client	OpenAI Python SDK	2.21.0+
Trends Analysis	pytrends	4.9.2+
Demo Server	Node.js HTTP (native)	—

4.1 Environment & Configuration

Configuration is managed through environment variables (dotenv). The `.env` file stores API keys for each LLM provider (DeepSeek, OpenAI GPT, Google Gemini, Mistral) as well as endpoints for the Python microservices and the demo blog upload API. Vite exposes these to the frontend via the `VITE_` prefix.

5. AI / LLM Integrations

CORE's core differentiator is its multi-LLM querying engine. The platform integrates with five major AI providers to deliver comprehensive brand visibility analysis:

Provider	Model(s)	API Format	Use Cases
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DeepSeek	deepseek-chat	OpenAI-compatible	Question generation, competitor analysis, article generation
OpenAI	GPT-4o, GPT-4o-mini	OpenAI native	Brand analysis, article generation, company context extraction
Google Gemini	2.5 Flash Lite, 2.5 Pro	Google REST API	Brand mention tracking, sentiment analysis
Anthropic	Claude 3.5 Sonnet	Anthropic Messages	Brand analysis (configurable)
Mistral	Mistral Large	OpenAI-compatible	Brand analysis (configurable)

5.1 Query Execution Pipeline

When an analysis is triggered, CORE sends each user-defined query to all selected LLM models in parallel. A rate-limiting mechanism introduces 2-second delays between batches to respect provider limits (especially Google Gemini). Responses are parsed for brand and competitor mentions, sentiment keywords, ranking positions within numbered lists, and cited URLs. Results are aggregated per model and per query for comparative analysis.

5.2 AI-Powered Features

- **Automatic Question Generation:** DeepSeek generates industry-relevant questions that potential customers would ask AI assistants.
- **Competitor Suggestion:** DeepSeek analyzes the brand context and suggests relevant competitors to track.
- **Company Context Extraction:** GPT-4o-mini combined with Selenium scraping summarizes a company's business from its website.
- **Trendy Keyword Discovery:** Google Trends API identifies trending keywords relevant to the brand's industry.
- **Mass Article Generation:** GPT-4o generates bulk blog articles optimized for AI search visibility.

6. Key Features

6.1 Brand Dashboard

The central dashboard provides a unified view of all brand metrics. Users can manage multiple brands and projects within workspaces, define custom queries, select target LLM models, and execute analyses. Results are displayed with mention counts, sentiment indicators, rank positions, and trend comparisons across models.

6.2 Geographic Intelligence (GEO)

CORE supports country and language targeting for analyses, enabling businesses to understand how their brand is perceived in different markets. Users can configure geographic parameters during project setup and compare regional differences in AI brand visibility.

6.3 Competitive Analysis

Beyond tracking the user's own brand, CORE monitors competitor mentions in the same AI responses. This provides comparative metrics showing who else is being recommended by AI assistants, how frequently they appear, and where opportunities exist to improve the user's brand visibility.

6.4 AI Article Engine

The Article Engine is a content generation module within the dashboard that automates the creation of blog articles. It fetches company context, discovers trending keywords, and uses GPT-4o to produce multiple articles in bulk. Articles can be published directly to external blog endpoints via HTTP POST, enabling a closed-loop workflow from insight to content to improved AI visibility.

6.5 Automated Monitoring

Users can schedule recurring analysis scans (daily, weekly, or custom intervals) to continuously track brand performance. The system maintains a history of analysis runs with timestamps, statuses, and result counts, allowing users to observe trends over time.

7. Data Layer & Storage

In its current MVP implementation, CORE uses a **localStorage-based database** (defined in `src/utlis/db.js`) for all data persistence. This approach was chosen to enable a fully client-side deployment without requiring backend infrastructure for the hackathon prototype. The data layer manages the following entities:

- **Users** — Account information and authentication state.
- **Workspaces** — Organizational containers for brands and projects.
- **Brands** — Brand profiles with name, industry, and description.
- **Competitors** — Tracked competitor entries linked to brands.
- **Projects** — Analysis projects with model selection and geographic targeting.
- **Queries** — User-defined or AI-generated questions for LLM analysis.
- **Results** — Parsed analysis outputs including mentions, sentiment, rank, and URLs.
- **Analysis Runs** — Execution metadata with timestamps, status, and result counts.

For production deployment, this layer would be replaced with a proper backend API and database (e.g., Firebase Firestore, PostgreSQL, or Supabase).

8. Python Backend Services

Two Python scripts provide backend capabilities that supplement the frontend application:

8.1 Company Context Extraction (`get_context.py`)

This service accepts a company URL, uses **Selenium WebDriver** to scrape the website content, and then sends the extracted text to **GPT-4o-mini** for summarization. The result is a structured summary of the company's business activity, products, and market positioning — which is then used by the AI

Article Engine and question generation features.

8.2 Trendy Keywords Discovery (`get_trendy_keywords.py`)

This service uses the **pytrends** library to query the Google Trends API for trending search terms related to the brand's industry. The discovered keywords inform both the question generation process and the article generation engine, ensuring content aligns with current search trends.

9. Demo Blog SaaS Module

CORE includes a standalone **demo-blog-saas** module — a lightweight Node.js HTTP server that simulates an external blog platform. It exposes REST endpoints for article upload (POST), retrieval (GET), and deletion (DELETE). Articles are stored as JSON files on disk. This module serves as a proof-of-concept for the end-to-end workflow: generate articles in CORE → publish to an external blog → verify content appears on the live blog page (<http://localhost:8787/blog>).

10. Future Vision

CORE's long-term vision is to evolve into a full **GEO Blog Engine Optimization** platform. The roadmap includes:

- **Mass Content Creation at Scale:** Automated generation of hundreds of optimized blog posts per campaign, tailored for specific industries and geographies.
- **LLM Search Optimization:** Strategies and tooling to improve how brands appear in AI models that can search the internet (Perplexity, ChatGPT with browsing, Gemini with Search).
- **Strategic Content Distribution:** Integration with CMS platforms, social media schedulers, and publishing networks for one-click content deployment.
- **Advanced Analytics:** Longitudinal tracking with trend visualization, ROI measurement for content campaigns, and predictive modeling of AI visibility.
- **Backend Migration:** Moving from `localStorage` to a production-ready backend with Firebase/Supabase, user authentication, and multi-tenant support.
- **Real-Time Monitoring:** Webhook-based alerting when brand visibility changes significantly across any monitored AI platform.

This report was generated for the CORE project, developed for the Google Hackathon 2026. For questions or collaboration inquiries, please refer to the project repository.